

Delta-Hedging an Up-and-Out Call

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1. Introduction

Delta-hedging a barrier option presents structural challenges absent from vanilla replication: the delta of an up-and-out call (UOC) becomes unstable as the underlying approaches the knock-out barrier H , generating disproportionate rebalancing costs and heavy-tailed hedging cost distributions. Using Monte Carlo simulation (300 trajectories, 252 rebalancing steps), we assess the performance of dynamic delta-hedging (DH) and compare it to the static replication strategy of [Derman, Ergener & Kani \(DEK, 1995\)](#) across a range of moneyness and maturity scenarios. All strategies are evaluated on common random numbers to ensure comparability.

2. Methodology

Hedging cost. We hold a short position on the UOC. The net hedging cost equals the portfolio construction cost minus the terminal portfolio value, plus the payoff, minus the premium. A positive cost denotes a net loss for the hedger. The primary metric used is the Hull Ratio.

Parameters. $S_0 = 50$, $K = 50$, $H = 60$, $r = 10\%$, $\sigma = 30\%$, $T = 0.75$, $q = 0$, $\varepsilon = 0.01$, 100 vanilla calls for the static portfolio. The underlying follows risk-neutral GBM and the barrier is monitored discretely at each step.

3. Dynamic Delta-Hedging Results

Distribution of hedging costs

We simulate a dynamic delta-hedging strategy on the 300 paths simulated with Monte Carlo method. The main results are reported in Table 1.

Table 1: Dynamic Hedging Results (Base Case)

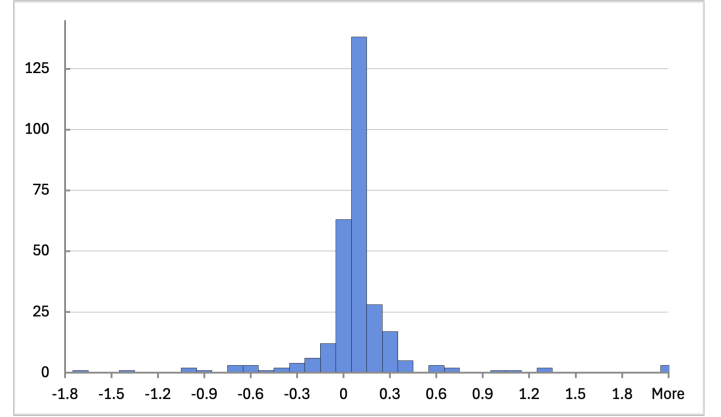
Metric	Value
Theoretical Price	0.3136
Mean Hedging Cost	0.0485
Std Deviation	0.5246
Hull Ratio	1.673
VaR 95% / CVaR 95%	0.3437 / 1.3453
Skewness / Kurtosis	8.17 / 104.96
% Loss > 0 / Max Loss	66.7% / 7.02

The distribution exhibits very high right skewness (8.17) and excess kurtosis (104.96), confirming that right tail risk dominates the hedging cost profile. This is further demonstrated by the CVaR 95%, which reaches 429% of the option price. Discrete rebalancing introduces a tracking error between the theoretical hedge and its implementation. [Boyle & Emanuel \(1980\)](#) show that this error decreases as $n^{-1/2}$, generating the non-normality features observed in the distribution. The hedging cost distribution is plotted in Figure 1.

KO vs. Alive decomposition and near-barrier costs

An important finding lies in the variance decomposition. Tail risk originates from surviving ITM paths (Std = 0.598), and not from knocked-out trajectories (Std = 0.341). Upon knock-out, the hedger retains the premium without paying a payoff — generating a predictable outcome. By contrast, ITM survivors require large payoff settlements compounded by frequent near-barrier rebalancing. To measure, this, we define the rebalancing cost per step as $c_j = |\Delta_j - \Delta_{j-1}| \cdot S_j$. Near-barrier steps ($S > 0.9H$) cost on average 0.815 versus 0.279 overall, representing a $2.92\times$ ratio. This is in line with academic literature, which identifies near-KO transaction costs as the primary source of barrier hedging inefficiency.

Figure 1: DH Hedging Cost Distribution

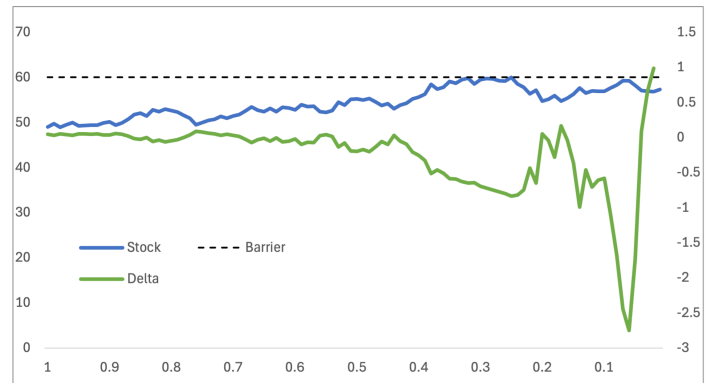


4. Sensitivity: Moneyness & Maturity

Delta dynamics near the barrier

The Delta of a UOC is composed of the option delta minus the call-in delta. As $S \rightarrow H$, the knock-out probability grows faster than the delta: the barrier effect dominates and the delta turns negative. Figure 2 illustrates this. Near expiry, this instability amplifies and the UOC delta collapses toward -2 in the final steps, mechanically forcing large rebalancing transactions from small price increments.

Figure 2: Stock Price and ATM UOC Delta



Hull Ratio, Mean Cost and VaR across scenarios

To understand the hardest case for dynamic hedging, we study the Hull Ratio, Mean Cost and VaR 95% across different scenarios. Results are reported in Table 2 and Table 3. The complete matrices are reported in the attached Excel file.

Table 2: Hull Ratio by Moneyness and Maturity

S/K	$T = 0.25$	$T = 0.50$	$T = 0.75$
0.80	0.856	2.645	1.978
0.90	0.611	1.205	2.302
1.00	0.923	0.993	2.154
1.10	1.024	1.534	3.258

The worst case ($S/K = 1.10$, $T = 0.75$, Hull Ratio = 3.26) combines near-barrier exposure with a long time horizon, maximising both delta instability and cumulative rebalancing costs. VaR 95% matrix provides another view: the near-barrier short-dated scenarios ($S/K = 1.00$, $T = 0.25$, VaR = 466% of C_0) exhibit the larger tail losses. Although long-dated options generate larger average replication uncertainty, options near expiry are exposed to more extreme hedging losses due to the very unstable gamma and the “jump” behaviour of the option value close to the barrier.

Table 3: VaR 95% by Moneyness and Maturity

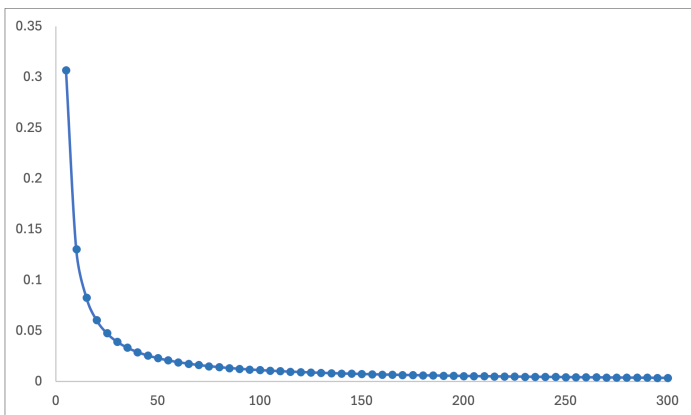
S/K	$T = 0.25$	$T = 0.50$	$T = 0.75$
0.80	0.185	0.607	0.861
0.90	0.559	0.591	0.999
1.00	1.463	0.745	1.067
1.10	1.426	0.603	0.625

5. Dynamic vs. Static Hedging

Static replication and convergence

The DEK replication portfolio builds at $t = 0$ a basket of N vanilla calls: one long call, and $N - 1$ short call expiring at intermediary maturities. This allow to cancel the UOC value on the barrier at discrete dates. The portfolio requires no rebalancing. Replication error, defined as the gap between the portfolio cost and the theoretical price, decreases with N : 41.7% at $N = 10$, 7.4% at $N = 50$, and 3.6% at $N = 100$.

Figure 3: Static Portfolio Replication Convergence



Within Black-Scholes, the static hedging cost is constant, making Hull Ratio and VaR uninformative. The relevant

metric is the replication error, which captures the mispricing due to the discrete number of short calls.

Replication error comparison

Table 4 compares the replication error of both strategies. For a limited set of 10 short calls, the static strategy is less precise than the dynamic one. For larger – and arguably unrealistic – set, the static error improves and outperforms dynamic.

Table 4: DH vs. SH Across Replication Frequencies

Metric	DH	SH(10)	SH(50)	SH(100)
Error	22.6%	41.7%	7.4%	3.6%

Comparison across scenarios

To compare strategies, we compute DH and SH hedging cost differences over a grid of moneyness and maturity. A positive entry means DH is more costly than SH, and vice versa.

Table 5: DH vs. SH Hedging Cost ($N = 10$)

S/K	$T = 0.25$	$T = 0.50$	$T = 0.75$
0.80	-0.018	-0.004	-0.052
0.90	-0.052	-0.153	-0.027
1.00	+0.014	-0.054	+0.004
1.10	+0.077	+0.005	+0.037

At $N = 10$, DH outperforms: the large SH replication error (41.7%) makes the static portfolio more costly than the dynamic hedge. The regime shifts sharply at $N = 50$: SH outperforms largely for ATM and ITM scenarios, where delta instability drives DH costs disproportionately higher.

Table 6: DH vs. SH Hedging Cost ($N = 50$)

S/K	$T = 0.25$	$T = 0.50$	$T = 0.75$
0.80	-0.006	+0.051	+0.034
0.90	+0.001	-0.051	+0.087
1.00	+0.123	+0.063	+0.111
1.10	+0.181	+0.088	+0.106

DH retains an advantage only for OTM short-dated options where rebalancing is infrequent. Results at $N = 100$ follow the same pattern as $N = 50$ with marginal improvements, as expected from the convergence analysis. This transition is consistent with Engelmann et al. (2006), who document empirically that static hedging dominates dynamic hedging within BS once the replication portfolio is highly granular.

6. Conclusion

Dynamic delta-hedging a UOC generates heavy-tailed costs driven by near-barrier delta instability: rebalancing costs are $2.92\times$ higher near H , and the Hull Ratio reaches 3.26 for ITM long-dated options. Static replication eliminates this variance entirely at the cost of a systematic replication error converging to 3.6% at $N = 100$. For $N \geq 50$, static hedging outperforms dynamic hedging across the majority of moneyness-maturity scenarios. However, as Nalholm & Poulsen (2006) showed, this advantage is contingent on model fidelity: outside Black-Scholes, the static portfolio accumulates model risk in ways that continuous rebalancing can partially correct.